Week 5 | **Exercises** p. 247 {6, 8, 16, 40} | **Assessed** Midterm + Final | **File** 2 / 11

4-1 Sample Spaces and Probability

فضاءات العينة والاحتمال

The whole of probability is one idea executed carefully: **name every outcome that could happen, name the ones you care about, compare the two counts.** Everything in §4-1 is machinery for doing that without losing an outcome.

1 · The five words you must define on command

These are CLO1 marks. They are asked as short definition questions and they are asked as the first part of long questions. Learn them as sentences, not as feelings.

TERM	DEFINITION (EXAM WORDING)	EXAMPLE
Probability experiment التجربة الاحتمالية	A chance process that leads to well-defined results called outcomes.	Rolling two dice.
Outcome النتيجة	The result of a *single trial* of a probability experiment.	Getting (5, 4).
Sample space `S` فضاء العينة	The set of *all possible* outcomes of a probability experiment. The number of elements in `S` is `n(S)`.	`S` = all 36 dice pairs.
Event الحدث	A set of outcomes of a probability experiment. Denoted `A, B, C…` or $E_1, E_2, E_3 \ldots$	`E` = "sum is 7".
Simple event الحدث البسيط	An event with **one** outcome.	"All three children are boys" = {BBB}.
Compound event الحدث المركّب	An event consisting of **two or more** outcomes.	"Exactly two girls" = {BGG, GBG, GGB}.

تمييز سريع

Outcome = نتيجة تجربة واحدة. **Sample space** = كل النتائج الممكنة. **Event** = مجموعة جزئية من فضاء العينة. إذا كانت تحتوي عنصرًا واحدًا فهي **simple**، وإذا احتوت عنصرين فأكثر فهي **compound**.

2 · Writing sample spaces — the three standard devices

الشرح بالعربي — كيف أكتب فضاء العينة؟

قبل حساب أي احتمال يجب أن تكتب **فضاء العينة كاملًا** دون أن تنسى أي نتيجة، لأن المقام في كل قانون هو $n(S)$. توجد ثلاث طرق:

1. **الجدول:** مناسب لرمي حجري نرد — $6 \times 6 = 36$ نتيجة على شكل أزواج مرتبة.
2. **السرد المنظّم:** تكتب النتائج بترتيب ثابت (كل الأولاد أولًا، ثم بنت واحدة في كل موقع، ثم بنتان...) حتى لا تسقط نتيجة.
3. **شجرة الاحتمالات:** كل مرحلة تتفرع إلى احتمالاتها، وعدد الأطراف النهائية = عدد النتائج. الشجرة هي **الدليل** على أنك لم تنسَ شيئًا.

DEVICE 1 — THE GRID (TWO DICE)

EXAMPLE 4-1 · ROLLING DICE

Find the sample space for rolling two dice.

Each die has 6 faces, and the dice are distinguishable (one red, one white). Lay them out as a 6 × 6 grid of ordered pairs:

$$S = \{(1,1),\ (1,2),\ \ldots,\ (1,6),\ (2,1),\ \ldots,\ (6,6)\}$$

So $n(S) = 36$

Why it matters: every dice question in the chapter divides by 36. Memorise the sum distribution now — sum 7 occurs 6 times, sums 6 and 8 occur 5 times each, 5 and 9 four times, 4 and 10 three times, 3 and 11 twice, 2 and 12 once. Total 36. ✔

DEVICE 2 — SYSTEMATIC LISTING (GENDER OF CHILDREN)

EXAMPLE 4-3 · GENDER OF CHILDREN

Find the sample space for the gender of the children if a family has three children. Use B for boy and G for girl.

$$S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$$

$n(S) = 8$ (= 2 × 2 × 2, by the Fundamental Counting Rule of §4-4)

Note the ordering discipline: all-boys first, then one girl in each of three positions, then two girls in each of three positions, then all girls. Listing in a fixed order is how you avoid dropping an outcome under exam pressure.

DEVICE 3 — THE TREE DIAGRAM

A **tree diagram** is a device consisting of line segments emanating from a starting point and also from the outcome point. It is used to determine all possible outcomes of a probability experiment.

EXAMPLE 4-4 · GENDER OF CHILDREN

Use a tree diagram to find the sample space for the gender of three children in a family.

Branch on the first child (B / G); from each of those, branch on the second child; from each of those, branch on the third. You get 2 × 2 × 2 = 8 terminal branches, read left to right:

FIRST CHILD	SECOND CHILD	THIRD CHILD	OUTCOME
B	B	B	BBB
B	B	G	BBG
B	G	B	BGB
B	G	G	BGG
G	B	B	GBB
G	B	G	GBG
G	G	B	GGB
G	G	G	GGG

Same eight outcomes as Example 4-3 — the tree is the *proof* that nothing was missed.

3 · Venn diagrams and set operations

A **Venn diagram** is a graphical way of representing the relationships between sets. The rectangle is S; circles inside it are events. Three standard configurations: overlapping ($A \cap B \neq \varnothing$), disjoint ($A \cap B = \varnothing$), and nested ($B \subset A$).

THE THREE SET OPERATIONS

Union $A \cup B$ — read "**A or B**" — all outcomes in either event *or both*

Intersection $A \cap B$ — read "**A and B**" — all outcomes in *both* events.

Complement $\bar{A}$ or A^c — read "**not A**" — all outcomes in S that are *no*

In probability, "or" is always *inclusive*: A or B or both. This is not the everyday English "or". It is the single most common source of lost marks in §4-2.

الشرح بالعربي — مخطط فن والعمليات على الأحداث

مخطط فن (Venn diagram): مستطيل يمثل فضاء العينة S ، وبداخله دوائر تمثل الأحداث.

- **الاتحاد** $A \cup B$ ويُقرأ **A or B**: كل العناصر الموجودة في A أو في B **أو في الاثنين معًا**.
- **التقاطع** $A \cap B$ ويُقرأ **A and B**: العناصر **المشتركة فقط** بين الحدثين.
- **المتمّمة** $\bar{A}$ أو A^c وتُقرأ **not A**: كل عناصر S التي **ليست** في A .

تنبيه: كلمة **or** في الاحتمال تعني «أحدهما أو كلاهما» دائمًا، وليست «أحدهما فقط» كما في الكلام اليومي. والمتمّمة تُحسب من داخل S فقط — في المثال أعلاه الرقم 8 يدخل في $\bar{A}$ لأنه ليس من عوامل 12.

WORKED EXAMPLE · 12-SIDED DIE

Roll a 12-sided die once. S = {1,…,12}. Let A = factors of 12 and B = factors of 8. Represent S, A, B on a Venn diagram, then find (a) $A \cup B$, (b) $A \cap B$, (c) $\bar{A}$, (d) $(A \cup B)^c$.

First, build the sets. Factors of 12: 1, 2, 3, 4, 6, 12 → $A = \{1, 2, 3, 4, 6, 12\}$. Factors of 8: 1, 2, 4, 8 → $B = \{1, 2, 4, 8\}$.

Venn layout. Inside A only: 3, 6, 12. Inside both: 1, 2, 4. Inside B only: 8. Outside both, still in S: 5, 7, 9, 10, 11.

1. (a) $A \cup B$ = {1, 2, 3, 4, 6, 8, 12} → **n = 7**
2. (b) $A \cap B$ = {1, 2, 4} → **n = 3**
3. (c) $\bar{A}$ = {5, 7, 8, 9, 10, 11} → **n = 6** (everything in S not in A — note 8 *is* included)
4. (d) $(A \cup B)^c$ = {5, 7, 9, 10, 11} → **n = 5**

Check: n(A ∪ B) + n((A ∪ B)^c) = 7 + 5 = 12 = n(S). ✔ Always run this check — it catches a dropped element in two seconds.

TRAP — THE COMPLEMENT MUST STAY INSIDE S

$\bar{A}$ is *not* "everything else in the universe". It is everything in **this sample space** that is not in A. In the example above, students routinely forget 8 belongs in $\bar{A}$ because it "feels like it belongs to B". B is irrelevant to $\bar{A}$.

4 · The three interpretations of probability

Probability is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur. The probability that event A occurs is written $P(A)$. There are three

basic interpretations:

1 · Classical

Uses sample spaces. Assumes **all outcomes are equally likely**. Computed before any data is collected — pure counting.

2 · Empirical (relative frequency)

Based on **observation**. Computed from a frequency distribution of data already collected.

3 · Subjective

A value based on an **educated guess or estimate**, employing opinions and inexact information.

الشرح بالعربي — أنواع الاحتمال الثلاثة

- **الكلاسيكي (Classical):** يُحسب من فضاء العينة **قبل** إجراء أي تجربة، بشرط أن تكون كل النتائج متساوية الفرص. مثال: احتمال ظهور 3 عند رمي نرد = 1/6 .
- **التجريبي (Empirical):** يُحسب من **بيانات جُمعت فعلًا** — جدول تكراري أو استبيان — ويساوي التكرار ÷ المجموع.
- **الشخصي (Subjective):** تقدير مبني على **رأي أو خبرة** شخص، بلا عدّ وبلا بيانات.

طريقة التمييز في الامتحان: فضاء عينة يُعدّ ← كلاسيكي. جدول أو «في استبيان شمل n شخصًا» ← تجريبي. رأي خبير ← شخصي.

CLASSICAL PROBABILITY

Equally likely events are events that have the same probability of occurring. The classical probability of any event `E` of a sample space `S` is:

CLASSICAL PROBABILITY

$$P(E) = \frac{n(E)}{n(S)}$$

`n(E)` = the number of outcomes in E. `n(S)` = the number of outcomes in S.

EXAMPLE 4-6 · GENDER OF CHILDREN

If a family has three children, find the probability that exactly two of the three children are girls.

1. **Sample space:** S = {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG}, so n(S) = 8.
2. **Event:** E = "exactly two of the three children are girls" = {BGG, GBG, GGB}, so n(E) = 3.
3. **Divide:** P(E) = n(E)/n(S) = 3/8 = **0.375**

Important (the lecturer flags this explicitly): sample spaces and events must be *written out using set notation* as shown above. Marks are allocated to the set, not only to the final number. Writing just "3/8" loses method marks.

THE FOUR PROBABILITY RULES

PROBABILITY RULES — LET EVENT E BE IN SAMPLE SPACE S

(1) $0 \le P(E) \le 1$

(2) If $P(E) = 0$, then E **cannot happen** (impossible).

(3) If $P(E) = 1$, then E is **certain**.

(4) The sum of the probabilities of all outcomes in S equals 1.

The probability line: 0 = Impossible · ~0.25 = Unlikely · 0.5 = 50–50 · ~0.75 = Likely · 1 = Certain.

الشرح بالعربي — القانون الكلاسيكي وقواعد الاحتمال

الاحتمال الكلاسيكي = عدد النتائج التي تحقق الحدث ÷ عدد كل النتائج الممكنة: $P(E) = n(E) / n(S)$.

قواعد الاحتمال الأربع:

1. الاحتمال دائمًا بين الصفر والواحد: $0 \le P(E) \le 1$.
2. إذا كان $P(E) = 0$ فالحدث **مستحيل** (مثل ظهور 9 على نرد عادي).
3. إذا كان $P(E) = 1$ فالحدث **مؤكد** (مثل ظهور عدد أقل من 7).
4. مجموع احتمالات كل النتائج في فضاء العينة = 1 .

استخدم القاعدة الأولى لفحص كل إجابة: أي احتمال **سالب** أو **أكبر من 1** يعني أن في الحل خطأ. وتذكّر أن الدكتور يطلب كتابة فضاء العينة والحدث **بصيغة المجموعات** {...} — عليها درجات.

EXAMPLES 4-8 & 4-9 · ROLLING A DIE

(a) When a single die is rolled, find the probability of getting a 9. (b) When a single die is rolled, what is the probability of getting a number less than 7?

(a) S = {1, 2, 3, 4, 5, 6}; E: getting a 9. It is impossible to get a 9, so E = { } and n(E) = 0.

$$P(E) = 0/6 = \boxed{0}$$

(b) S = {1, 2, 3, 4, 5, 6}; E: getting a number less than 7. Every face qualifies, so E = {1, 2, 3, 4, 5, 6} and n(E) = 6.

$$P(E) = 6/6 = \boxed{1} \text{ — the event is } \textbf{certain}.$$

These two examples exist to demonstrate rules (2) and (3). An exam question that produces `P = 0` or `P = 1` is testing whether you *recognise* impossibility and certainty — say the words in your answer.

5 · Complementary events

The **complement of an event E** is the set of outcomes in the sample space that are not included in the outcomes of event E. It is denoted $\bar{E}$ (read "E bar").

RULE FOR COMPLEMENTARY EVENTS

$$P(\bar{E}) = 1 - P(E) \quad \text{or} \quad P(E) = 1 - P(\bar{E}) \quad \text{or} \quad P(E) + P(\bar{E}) = 1$$

This is the highest-leverage formula in the chapter. Whenever a question contains "**not**", "**at least one**", "**neither**", or "**none**", reach for it before you do anything else.

الشرح بالعربي — المتمّمة

متمّمة الحدث E هي كل النتائج التي **لا تحقق** E. وبما أن الحدث إما أن يقع أو لا يقع، فمجموع الاحتمالين = 1، ومنه:

$$P(\bar{E}) = 1 - P(E)$$

متى تستخدمها؟ عندما ترى في السؤال: **not** (ليس)، **neither ... nor** (لا هذا ولا ذاك)، **none** (ولا واحد)، **at least one** (واحد على الأقل). هذه القاعدة كثيرًا ما تحوّل سؤالًا طويلًا إلى عملية طرح واحدة. مثال: 23% يفضّلون الفانيلا، إذن احتمال **ليس** الفانيلا = $1 - 0.23 = 0.77$.

EXAMPLE 4-10 · FINDING COMPLEMENTS

Find the complement of each event: (a) selecting a month that has 30 days; (b) selecting a day of the week that begins with the letter S; (c) rolling two dice and getting a number whose sum is 7.

1. **(a)** Months have 28, 30 or 31 days. Complement = **selecting a month that has 28 or 31 days.**
2. **(b)** Days starting with S are Saturday and Sunday. Complement = **selecting a day that does not begin with the letter S** — that is, Monday, Tuesday, Wednesday, Thursday or Friday.
3. **(c)** The possible sums of two dice are 2 through 12. Complement = **getting a sum of 2, 3, 4, 5, 6, 8, 9, 10, 11, or 12.**

NOTE ON THE SLIDE

The lecture slide for part (b) prints "does not begin with the letter **T**", which is a typo carried from an earlier version of the example. The event is "begins with S", so its complement is "does *not* begin with S". Write S in your answer and you are correct against the textbook.

EXAMPLE 4-11 · FAVORITE ICE CREAM FLAVORS

In a study, 23% of the people surveyed said that vanilla was their favorite flavor of ice cream. If a person is selected at random, find the probability that the person's favorite flavor is *not* vanilla.

Let E be "favorite flavor is vanilla", with P(E) = 0.23.

$$P(\text{not vanilla}) = 1 - P(E) = 1 - 0.23 = \boxed{0.77} \quad (\text{or } 77\%)$$

6 · Empirical probability

Given a frequency distribution, the probability of an event being in a given class is the relative frequency of that class:

EMPIRICAL (RELATIVE FREQUENCY) PROBABILITY

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

`f` = frequency of the class (event); `n` = the sample size. Based on **observation**, not on assuming equally likely outcomes.

الشرح بالعربي — الاحتمال التجريبي

يُحسب من البيانات المجموعة: `P(E) = f / n` ، حيث `f` تكرار الفئة و `n` حجم العينة الكلي. مثال: سألت 50 شخصًا وأجاب 3 منهم «قطار أو حافلة»، فالاحتمال `3/50 = 0.06` .

نصيحة: اجمع عمود التكرارات وتأكد أنه يساوي `n` ، واجمع عمود الاحتمالات وتأكد أنه يساوي `1` . وإذا طُلب احتمال فئتين معًا (A أو B) فاجمع تكراريهما ثم اقسم.

EXAMPLE 4-12 · TRAVEL SURVEY

A researcher asked 50 people who plan to travel over the Thanksgiving holiday how they will get to their destination: Drive 41, Fly 6, Train or bus 3. Find the probability that a person will travel by train or bus.

METHOD	FREQUENCY F	EMPIRICAL PROBABILITY F/N
Drive	41	41/50 = 0.82
Fly	6	6/50 = 0.12
Train or bus	3	3/50 = 0.06
Total	**50**	**50/50 = 1.00**

`P(train or bus) = f/n = 3/50 =` **0.06**

The total column summing to exactly 1.00 is probability rule (4) in action — use it as a check on every frequency table you build.

EXAMPLE 4-13 · DISTRIBUTION OF BLOOD TYPES

In a sample of 50 people, 21 had type O blood, 22 had type A, 5 had type B, and 2 had type AB. Set up a frequency distribution and find: (a) P(type O); (b) P(type A or type B); (c) P(neither type A nor type O); (d) P(not type AB).

TYPE	FREQUENCY
A	22
B	5
AB	2
O	21
Total	**50**

1. **(a)** P(O) = f/n = 21/50 = **0.42**
2. **(b)** P(A or B) = 22/50 + 5/50 = 27/50 = **0.54** (add the frequencies of the two classes — the types are mutually exclusive)
3. **(c)** "Neither A nor O" means the person has type B or type AB: P = 5/50 + 2/50 = 7/50 = **0.14**
4. **(d)** P(not AB) = 1 − P(AB) = 1 − 2/50 = 48/50 = **0.96**

Read the structure: (b) is the addition rule, (c) is a translated complement ("neither…nor" = complement of a union), (d) is the complement rule. One table, three different tools. This is exactly how a long exam question is built.

EXAMPLE 4-14 · SLEEP HOURS

A survey found the following distribution for hours of sleep per night: Less than 6 h → 12; 6 h → 26; 7 h → 30; 8 h → 28; More than 8 h → 4. Total 100. Find, for a person selected at random: (a) sleeps 8 hours; (b) sleeps fewer than 7 hours; (c) sleeps **at most** 8 hours; (d) sleeps **at least** 7 hours.

1. **(a) Exactly 8 hours** — one class only: P = 28/100 = **0.28**
2. **(b) Fewer than 7 hours** — strictly below 7, so "less than 6" and "6": P = (12 + 26)/100 = 38/100 = **0.38**
3. **(c) At most 8 hours** — 8 and everything below it: P = (12 + 26 + 30 + 28)/100 = 96/100 = **0.96**
   Faster: 1 − P(more than 8) = 1 − 4/100 = 0.96 ✔
4. **(d) At least 7 hours** — 7 and everything above it: P = (30 + 28 + 4)/100 = 62/100 = **0.62**

7 · The five phrases — translate before you calculate

The lecturer gives this as a standalone "Important Note" slide, which is a strong signal that it is examined. Consider the values **3, 4, 5, 6, 7**:

PHRASE	VALUES USED	SYMBOL
Exactly 5	5	$X = 5$
Less (fewer) than 5	3, 4	$X < 5$ — 5 is *excluded*
At most 5 (5 or less)	3, 4, 5	$X \leq 5$ — 5 is *included*
More (greater) than 5	6, 7	$X > 5$ — 5 is *excluded*
At least 5	5, 6, 7	$X \geq 5$ — 5 is *included*

الترجمة الحرفية — احفظها

- **Exactly 5** = بالضبط 5 فقط.
- **Less / fewer than 5** = أقل من 5، **بدون** الـ 5.
- **At most 5** = 5 فأقل، **مع** الـ 5.
- **More / greater than 5** = أكثر من 5، **بدون** الـ 5.
- **At least 5** = 5 فأكثر، **مع** الـ 5.

القاعدة الذهبية: كلمة (at most / at least) **at** تعني أن الرقم نفسه **داخل** الحساب. وكلمة (less than / more than) **than** تعني أن الرقم نفسه **خارج** الحساب.

8 · Subjective probability

Subjective probability uses a probability value based on an educated guess or estimate, employing opinions and inexact information. No sample space is counted and no data is tabulated — a qualified person simply assigns a number.

- A sportswriter may say there is a **70% probability** that the Oman football team will win the Gulf Cup next year.
- A physician might say that, on the basis of her diagnosis, there is a **30% chance** the patient will need an operation.
- A seismologist might say there is an **80% probability** that an earthquake will occur in a certain area.

الشرح بالعربي — الاحتمال الشخصي

رقم يعطيه شخص بناءً على **خبرته أو رأيه**، مثل قول محلل رياضي إن احتمال فوز منتخب عُمان بكأس الخليج 70% ، أو قول طبيب إن احتمال حاجة المريض لعملية 30% . لا يوجد فضاء عينة يُعدّ ولا جدول بيانات — مجرد تقدير مدروس. غالبًا يأتي في الامتحان كسؤال: «ما نوع هذا الاحتمال؟»

HOW THIS IS EXAMINED

You will be shown a statement and asked *which type of probability is this?* The tell: counting a sample space → **classical**. A frequency table or "in a survey of n people" → **empirical**. An expert's opinion with no data behind it → **subjective**. One mark, two seconds, never miss it.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests	**01 · Lesson 4-1 — Sample Spaces and Probability** ← this file
02 · Lesson 4-2 — The Addition Rules	03 · Lesson 4-3 — Multiplication Rules and Conditional Probability
04 · Lesson 4-4 — Counting Rules	05 · Lesson 4-5 — Probability and Counting Rules
06 · Master formula sheet	07 · Which rule do I use?
08 · The 14 traps that cost marks	09 · Review exercises, solved
10 · The A+ study protocol	

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.